

2024, Mar. 2024 Release LOC:Acute Adult

Diabetic Ketoacidosis

Utilization Benchmarks: Multiple options available, see table below for options

Overview

Select Day

Initial review ^(1, 2)Episode Day 1 ⁽¹⁾Episode Day 2 ⁽¹⁸⁾Episode Day 3 ⁽⁴⁴⁾

Episode Day 4

Discharge Screens ⁽⁴⁶⁾

Utilization Benchmarks

Notes

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Overview

Introduction: Diabetic ketoacidosis (DKA) is a serious acute metabolic complication of diabetes mellitus characterized by the metabolic triad of hyperglycemia, anion gap metabolic acidosis, and ketosis. DKA occurs when there is a relative or absolute decrease in insulin levels and an excess of counterregulatory hormones. As a result of this imbalance, gluconeogenesis and glycogenolysis occur with increased glucose production and decreased peripheral glucose utilization. This subsequently leads to hyperglycemia, hyperosmolality, increased lipolysis, and ketogenesis, thereby causing ketosis and metabolic acidosis. Despite improvements in diabetes care, DKA remains a life-threatening condition that, in most cases, requires urgent hospitalization for monitoring and management to correct dehydration, ketosis, and acidosis (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Linder and Shill, Ferri's Clinical Advisor. 2021; French et al., Bmj 2019, 365: l1114; Fayfman et al., Med Clin North Am 2017, 101: 587-606).

Evaluation and Treatment:

Although most common in type 1 diabetes mellitus (DM) patients, diabetic ketoacidosis (DKA) can also occur in patients with type 2 DM. Common precipitating factors for DKA are infection and insulin noncompliance, and DKA can be the first presentation of DM. Stressful events (illness, trauma, surgery), certain medications, and substance use disorder can also result in DKA. Symptoms may include polyuria, polydipsia, vomiting, and abdominal pain, while physical examination can reveal signs of dehydration, ketotic breath (fruity smelling), Kussmaul breathing, abdominal tenderness, and neurologic symptoms (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; French et al., Bmj 2019, 365: l1114; Fayfman et al., Med Clin North Am 2017, 101: 587-606; Nyenwe and Kitabchi, Metabolism 2016, 65: 507-21).

The diagnosis of DKA is based on laboratory findings of elevated glucose greater than 250 mg/dL (13.9 mmol/L), ketonemia or ketonuria, and acidosis. According to the American Diabetes Association, there are three levels of severity of DKA, defined as (American Diabetes Association, Diabetes Care 2024, 47: S1-S314):

- Mild: venous pH 7.25-7.30 or serum bicarbonate 15-18 mEq/L (15-18 mmol/L), anion gap 10-12 mEq/L (10-12 mmol/L)
- Moderate: venous pH 7.00-7.24 or serum bicarbonate 10-14.9 mEq/L (10-14.9 mmol/L), anion gap > 12 mEq/L (12 mmol/L)
- Severe: venous pH < 7.0 or serum bicarbonate < 10 mEq/L (<10 mmol/L), anion gap > 12 mEq/L (12 mmol/L)

While most patients with DKA have significantly elevated glucose values, around 10% of patients may present with euglycemic diabetic ketoacidosis in which they have a lower blood glucose level than anticipated. The etiology of euglycemic ketoacidosis includes but is not limited to pregnancy, decreased caloric intake, alcohol use disorders,

chronic liver disease, insulin administration en route to the hospital, and sodium-glucose co-transporter 2 (SGLT2) inhibitors. Successful management and treatment of DKA requires addressing the patient's acidosis, ketosis, hyperosmolality, and hyperglycemia which involves monitoring and correcting electrolyte abnormalities as needed, intravenous fluid resuscitation, insulin administration, monitoring for cerebral edema, and identification and treatment of precipitating events. DKA is associated with many complications. Cerebral injury is a complication that primarily contributes to morbidity and mortality in DKA, with a mortality rate of 21%-24%.

High-quality diabetes care requires addressing barriers impeding disease management and providing diabetes self-management education and support (DSMES) that is specific to the needs of each individual patient. The times of diagnosis and when acute complications arise represent critical moments to evaluate knowledge deficits and needs and intervene with DSMES that should begin immediately and continue throughout the patient's journey through the care continuum. Education provided in a comprehensive and easy-to-understand format without unnecessary complexity can prevent the recurrence of DKA and other complications and improve patient self-management, health status, and clinical outcomes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Linder and Shill, Ferri's Clinical Advisor. 2021; French et al., Bmj 2019, 365: l1114; Dhatariya and Vellanki, Curr Diab Rep 2017, 17: 33; Fayfman et al., Med Clin North Am 2017, 101: 587-606).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽³⁾– Blood glucose > 250 mg/dL(13.9 mmol/L) ⁽⁴⁾– Ketones, elevated ⁽⁵⁾● **Acidosis, ≥ One:**

– Anion gap 10-12 mEq/L(10-12 mmol/L)

– Arterial or venous pH 7.25-7.30 ⁽⁶⁾– Serum HCO₃ or CO₂ 15-18 mEq/L(15-18 mmol/L) ⁽⁷⁾● **CRITICAL, One:** ⁽⁸⁾● **Finding, ≥ One:**● **Labs, All:** ⁽⁹⁾

– Blood glucose > 250 mg/dL(13.9 mmol/L)

– Ketones, elevated ⁽⁵⁾● **Acidosis, ≥ One:**

– Anion gap > 12 mEq/L(12 mmol/L)

– Arterial or venous pH < 7.25 ⁽⁶⁾– Serum HCO₃ or CO₂ < 15 mEq/L(15 mmol/L) ⁽⁷⁾– Coma, stupor, obtundation or GCS ≤ 8 ⁽¹⁰⁾– Pregnant individual ⁽¹¹⁾● **Episode Day 1, One:** ⁽¹⁾

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽³⁾– Blood glucose > 250 mg/dL(13.9 mmol/L) ⁽⁴⁾– Ketones, elevated ⁽⁵⁾● **Acidosis, ≥ One:**

– Anion gap 10-12 mEq/L(10-12 mmol/L)

– Arterial or venous pH 7.25-7.30 ⁽⁶⁾– Serum HCO₃ or CO₂ 15-18 mEq/L(15-18 mmol/L) ⁽⁷⁾● **Intervention, Both:** ⁽¹²⁾– Insulin ^(13, 14, 15)– IV fluid resuscitation ⁽¹⁶⁾● **CRITICAL, Both:** ⁽⁸⁾● **Finding, ≥ One:**● **Labs, All:** ⁽⁹⁾

– Blood glucose > 250 mg/dL(13.9 mmol/L)

– Ketones, elevated ⁽⁵⁾● **Acidosis, ≥ One:**

– Anion gap > 12 mEq/L(12 mmol/L)

– Arterial or venous pH < 7.25 ⁽⁶⁾– Serum HCO₃ or CO₂ < 15 mEq/L(15 mmol/L) ⁽⁷⁾– Coma, stupor, obtundation or GCS ≤ 8 ⁽¹⁰⁾– Pregnant individual ⁽¹¹⁾● **Intervention, Both:** ⁽¹²⁾– Insulin, continuous ^(13, 15, 17)– IV fluid resuscitation ⁽¹⁶⁾● **Episode Day 2, One:** ⁽¹⁸⁾● **OBSERVATION, One:**● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹⁹⁾– Blood glucose within acceptable limits ⁽²⁰⁾

- Lab values within acceptable limits ^(20, 21)
- Tolerating PO ⁽²²⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - **Blood glucose, All:**
 - **Finding, ≥ One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring ^(23, 24)
 - Insulin or oral glucose-lowering medication initiation, adjustment, or dose held ⁽²⁵⁾
 - **Hypokalemia, All:** ^(26, 27)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - No electrocardiogram (ECG) changes ^(28, 29)
 - Potassium repletion (includes PO)
 - **Management of precipitating cause of diabetic ketoacidosis, ≥ One:** ⁽³⁰⁾
 - Infection and anti-infective ≤ 24h
 - Pump failure actual or suspected and transition to insulin (SC only) or back to insulin pump ≤ 24h
 - Patient or caregiver knowledge deficit and diabetes education ^(31, 32)
- **ACUTE, One:** ⁽³³⁾
 - Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:** ⁽³⁴⁾
 - HFNC and respiratory monitoring every 3-4h ^(35, 36)
 - **Hypokalemia, All:** ^(26, 27)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - PVCs > 6/min
 - Potassium repletion
 - NIPPV ⁽³⁷⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽³⁸⁾
- **CRITICAL, ≥ One:**
 - Diabetic ketoacidosis and insulin, continuous ^(15, 17)
 - HFNC and respiratory monitoring every 1-2h ^(35, 36, 38)
 - **Hypokalemia, All:** ^(26, 27, 39)
 - Potassium < 3.0 mEq/L(3.0 mmol/L)
 - **ECG findings, ≥ One:** ⁽³⁹⁾
 - Multifocal PVCs
 - R on T phenomenon ⁽⁴⁰⁾
 - T and U wave fusion ⁽⁴¹⁾
 - U wave ⁽⁴²⁾
 - Potassium repletion ⁽²⁷⁾
 - Mechanical ventilation
 - **NIPPV and, ≥ One:** ⁽³⁷⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁴³⁾
- **Episode Day 3, One:** ⁽⁴⁴⁾
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹⁹⁾
 - Blood glucose within acceptable limits ⁽²⁰⁾
 - Lab values within acceptable limits ^(20, 21)
 - Patient or caregiver knowledgeable about disease and understands the regimen
 - Tolerating PO ⁽²²⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
 - For diabetic ketoacidosis use Episode Day 3 at a higher level of care
 - For a condition other than diabetic ketoacidosis use appropriate subset (Episode Day 1) based on clinical

findings

– Refer for secondary review

● **ACUTE, One:**

● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹⁹⁾

– Blood glucose within acceptable limits ⁽²⁰⁾

– Lab values within acceptable limits ^(20, 21)

– Tolerating PO ⁽²²⁾

– No complication or active comorbidity relevant to this episode of care ⁽⁴⁵⁾

● **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**

● **Blood glucose, All:**

● **Finding, ≥ One:**

– > 250 mg/dL(13.9 mmol/L)

– < 70 mg/dL(3.9 mmol/L)

– Blood glucose monitoring ^(23, 24)

– Insulin or oral glucose-lowering medication initiation, adjustment, or dose held ⁽²⁵⁾

– Patient or caregiver knowledge deficit and diabetes education ^(31, 32)

– Post Critical care ≤ 24h

● **INTERMEDIATE, ≥ One:**

– HFNC and respiratory monitoring every 3-4h ^(35, 36)

– NIPPV ⁽³⁷⁾

– Oxygen ≥ 40%(0.40), ≤ 2d ⁽³⁸⁾

● **CRITICAL, ≥ One:**

– Diabetic ketoacidosis and insulin, continuous ^(15, 17)

– HFNC and respiratory monitoring every 1-2h ^(35, 36, 38)

– Mechanical ventilation

● **NIPPV and, ≥ One:** ⁽³⁷⁾

– FiO₂ > 50%(0.50)

– Respiratory intervention every 1-2h ⁽⁴³⁾

● **Episode Day 4, One:**

● **ACUTE, One:**

● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹⁹⁾

– Blood glucose within acceptable limits ⁽²⁰⁾

– Lab values within acceptable limits ^(20, 21)

– Patient or caregiver knowledgeable about disease and understands the regimen

– Tolerating PO ⁽²²⁾

– No complication or active comorbidity relevant to this episode of care ⁽⁴⁵⁾

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**

– **Use Extended Stay subset**

– Refer for secondary review

● **INTERMEDIATE, One:**

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**

– **Use Extended Stay subset**

– Refer for secondary review

● **CRITICAL, One:**

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**

– **Use Extended Stay subset**

– Refer for secondary review

Discharge, Level of Care, One: ⁽⁴⁶⁾

● **HOME, Both:**

- Level of care appropriateness, **Both:**
 - Home environment safe and accessible ⁽⁴⁷⁾
 - Patient or caregiver demonstrates ability to manage care
- Diabetic ketoacidosis (DKA) discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁴⁸⁾
 - Education on the importance of, **All:**
 - Prevention and treatment of hypoglycemia
 - Home monitoring of blood glucose
 - Sick day management
 - Foot inspection and care
 - Pneumococcal and annual influenza vaccine
 - Annual eye examination
 - Lifestyle modification to prevent hypertension and dyslipidemia ⁽⁴⁹⁾
 - Smoking cessation, if indicated
 - Glycemic control and hypertension management, if hypertension present, to prevent retinopathy, nephropathy, and neuropathy ⁽⁵⁰⁾
 - Medication reconciliation ⁽⁵¹⁾
 - Follow-up care planned ⁽⁵²⁾
 - Identify and address transportation needs
- **HOME CARE, Both:**
 - Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽⁴⁷⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(53, 54)
 - Skilled services, ≥ **One:** ⁽⁵⁵⁾
 - Skilled nursing ⁽⁵⁶⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁵⁷⁾
 - DKA discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁴⁸⁾
 - Education on the importance of, **All:**
 - Prevention and treatment of hypoglycemia
 - Home monitoring of blood glucose
 - Sick day management
 - Foot inspection and care
 - Pneumococcal and annual influenza vaccine
 - Annual eye examination
 - Lifestyle modification to prevent hypertension and dyslipidemia ⁽⁴⁹⁾
 - Smoking cessation, if indicated
 - Glycemic control and hypertension management, if hypertension present, to prevent retinopathy, nephropathy, and neuropathy ⁽⁵⁰⁾
 - Medication reconciliation ⁽⁵¹⁾
 - Follow-up care planned ⁽⁵²⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽⁵⁸⁾
 - Skilled treatment, ≥ **One:**
 - Clinical assessment ⁽⁵⁹⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁶⁰⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 1\text{x/wk}$
- Treatment precluded in a lower level
- Services, $\geq$ **One**:
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁵⁶⁾
- Therapy, **All**:
 - Goal directed therapy with rehabilitation potential ^(61, 62)
 - Functional impairments requiring at least supervision ⁽⁶³⁾
 - Able to tolerate 1h/d of skilled therapy $\geq 5\text{d/wk}$
- Complete prior to facility transfer, **All**:
 - Medication reconciliation ⁽⁵¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both**:
 - Level of care appropriateness, **All**:
 - Medical practitioner, NP, PA assessment or oversight $\geq 2\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, $\geq$ **One**:
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁵⁶⁾
 - Therapy, **All**:
 - Goal directed therapy with rehabilitation potential ^(61, 62)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶³⁾
 - Able to tolerate 2-3h/d of skilled therapy $\geq 5\text{d/wk}$ and ≥ 1 therapy discipline
 - Complete prior to facility transfer, **All**:
 - Medication reconciliation ⁽⁵¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both**:
 - Level of care appropriateness, **All**:
 - Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **All**:
 - Goal directed therapy with rehabilitation potential ^(61, 62)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶³⁾
 - Able to tolerate 2-3h/d of skilled therapy $\geq 5\text{d/wk}$ and ≥ 2 therapy disciplines
 - Complete prior to facility transfer, **All**:
 - Medication reconciliation ⁽⁵¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both**:
 - Level of care appropriateness, **Both**:
 - Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$
 - Services, **All**:
 - Goal directed therapy with rehabilitation potential ^(61, 62)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶³⁾
 - Able to tolerate $\geq 15\text{h/wk}$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁶⁴⁾
 - Complete prior to facility transfer, **All**:
 - Medication reconciliation ⁽⁵¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

- **LONG TERM ACUTE CARE, Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁶⁵⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(66, 67)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁶⁸⁾
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁵¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
637 DIABETES WITH MCC	n/a	4.1	CMS GMLOS
638 DIABETES WITH CC	n/a	3	CMS GMLOS
639 DIABETES WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
DIABETIC KETOACIDOSIS	11%	3	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

The diagnostic criteria for mild diabetic ketoacidosis are a blood glucose level greater than 250 mg/dl (13.9 mmol/L), arterial pH 7.25-7.30, serum bicarbonate 15-18 mEq/L (15-18 mmol/L), and anion gap greater than 10 mEq/L (10 mmol/L) with ketonemia or ketonuria (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; French et al., Bmj 2019, 365: l1114).

5:

Diagnostic testing for elevated ketone levels includes evaluation of one or more of the following:

- Beta-hydroxybutyrate > 3 mmol/L (31 mg/dL)
- Serum ketones positive (ketonemia)
- Urine ketones positive (ketonuria) ≥ 2+ (moderate to large)

6:

Venous blood gases (VBG) may be used to assess a patient's ventilation or metabolic status. Venous pH or HCO_3 values closely approximate corresponding values derived from arterial blood gases (ABG) in conditions such as chronic obstructive pulmonary disease (COPD), respiratory distress syndrome, neonatal sepsis, renal failure, pneumonia, diabetic ketoacidosis, and status epilepticus (McKeever et al., Thorax 2016, 71: 210-5).

7:

Measurement of bicarbonate (HCO_3) in venous blood is done indirectly via a technique that measures carbon dioxide (CO_2). Total CO_2 may be slightly higher, up to 5%(0.05) above the actual HCO_3 , due to other substances in the plasma. Since the difference between total CO_2 and actual HCO_3 is small, the 2 quantities are usually treated as being equal.

8:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

9:

The diagnostic criteria for moderate to severe diabetic ketoacidosis are a blood glucose level greater than 250 mg/dL (13.9 mmol/L), arterial pH less than 7.25, serum bicarbonate less than 15 mEq/L (15 mmol/L), and anion gap greater than 12 (12 mmol/L), with ketonemia or ketonuria (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; French et al., Bmj 2019, 365: l1114).

10:

Glasgow Coma Scale (GCS) is a tool commonly used to measure neurological function. GCS evaluates the level of consciousness in acutely ill patients. A patient's GCS is the combined total score accumulated through the evaluation of the following 3 areas (ranging from 3-15) (Chou et al., In: Glasgow Coma Scale for Field Triage of Trauma: A Systematic Review. 2017):

- Eye opening - 4 = Spontaneous, 3 = To voice, 2 = To pain, 1 = None
- Verbal response - 5 = Oriented, 4 = Confused, 3 = Inappropriate, 2 = Incomprehensible sounds, 1 = None
- Motor response - 6 = Obeys verbal commands, 5 = Localizes to pain, 4 = Withdraws to pain, 3 = Flexion response to pain, 2 = Extension response to pain, 1 = None

11:

Pregnancy is a ketogenic state that increases the risk of diabetic ketoacidosis (DKA) for individuals with a history of type 1 or type 2 diabetes mellitus who become pregnant. DKA can manifest in a euglycemic state for pregnant individuals and potentially be unrecognized. DKA increases the risk of stillbirth and puts the mother and fetus at a higher risk of morbidity and mortality (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

12:

Diabetic ketoacidosis (DKA) treatment and management goals include restoring circulatory volume and tissue perfusion, correcting hyperglycemia, electrolyte imbalances, and acidosis, and addressing the underlying cause of DKA (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

13:

Instruction: Medication administration includes continuous infusion and titration of medications. For medications that are titrated, the rate and dose are adjusted based on clinical monitoring and laboratory results.

14:

Evidence has shown that there is no significant difference in outcomes for intravenous human regular insulin versus subcutaneous rapid-acting analogs while treating mild diabetic ketoacidosis (DKA). When subcutaneous insulin administration is used, it is essential to provide conventional DKA treatment that includes aggressive fluid replacement, frequent blood glucose monitoring, treatment of any concurrent illness or precipitating factors, and appropriate follow-up to avoid recurrent hyperglycemic crises (American Diabetes Association, Diabetes Care 2023, 46: S1-S298; Blonde et al., Endocr Pract 2022, 28: 923-1049; French et al., Bmj 2019, 365: l1114).

15:

Resolution of diabetic ketoacidosis (DKA) is demonstrated by blood glucose level less than 200 mg/dL (11.1 mmol/L) and two of the following:

- Anion gap ≤ 12 mEq/L (12 mmol/L) (lab value or calculated)
- Serum bicarbonate level > 15 mEq/L (15 mmol/L)
- Arterial or venous pH > 7.3

Basal insulin administration is recommended two to four hours prior to the cessation of intravenous insulin to prevent rebound hyperglycemia and ketoacidosis (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; French et al., Bmj 2019, 365: l1114).

16:

Administration of intravenous fluid is the first line of treatment for diabetic ketoacidosis (DKA) to restore intravascular volume, lower blood glucose, ensure adequate tissue perfusion and assist in resolving metabolic acidosis. A 0.9% sodium chloride solution is recommended for initial fluid resuscitation with an adjustment of the rate of fluid replacement after the initial hour based on the patient's hemodynamic, electrolyte, cardiac, renal, liver, and fluid status. As hyperglycemia improves, intravenous fluids with dextrose should be administered simultaneously with insulin until DKA resolves to avoid hypoglycemia (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; Blonde et al., Endocr Pract 2022, 28: 923-1049; French et al., Bmj 2019, 365: l1114).

17:

The administration of insulin via continuous intravenous infusion is the standard of care and the most effective method for achieving glycemic targets and closing the anion gap for critically ill individuals with diabetic ketoacidosis (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

18:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights. For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

19:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

20:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

21:

A glucose target between 100-180 mg/dL (5.6-10.0 mmol/L) is recommended for most patients in noncritical care units (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

22:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

23:

The target glycemic range for non-critically ill hospitalized patients with diabetes is 100-180 mg/dL (5.6-10.0 mmol/L). Blood glucose concentrations greater than 180 mg/dL (10.0 mmol/L) may be acceptable, at times, to reduce the risk of harmful outcomes in patients with severe comorbidities or decreased life expectancy. Blood glucose targets should be individualized and modified if there are identifiable risks for adverse effects such as a history of frequent hypoglycemia or hypoglycemia unawareness (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

24:

In non-critically ill hospitalized patients with adequate nutritional intake, recommendations are to complete point-of-care (POC) glucose testing immediately before meals. Alternatively, in patients with inadequate oral intake, oral intake restrictions, or those on continuous enteral or parenteral nutrition, recommendations are to complete POC glucose testing every 4-6 hours. However, blood glucose monitoring is individualized. Additional or more frequent blood glucose testing is necessary if the patient cannot effectively communicate their needs, recognize signs or symptoms of a complication, or has a history of increased risk of diabetes-related adverse events (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

25:

Instruction: Medication adjustment includes initiation of a new drug, change in dosage, and/or administration frequency of a medication. Blood glucose values guide medication dosing decisions and adjustments. Administration of sliding scale insulin must occur at least 3 times within 24 hours. A continuous infusion rate must have at least 2 rate adjustments within 24 hours (including insulin pump rate adjustments). A glucose-lowering medication dose withheld based on normal or low blood glucose levels is included in the count of doses

administered within the 24-hour timeframe.

26:

Hypokalemia rarely occurs in isolation and is usually secondary to a disease process or overuse of diuretics. Criteria should be reviewed for the underlying cause of the hypokalemia (e.g., diabetic ketoacidosis, gastroenteritis) to ensure the patient is managed appropriately.

27:

When a patient with hypokalemia is not responding to potassium repletion, the magnesium level should be checked since hypomagnesemia could be contributing to the refractory hypokalemia.

28:

In patients with no underlying heart disease, occasional, nonconsecutive premature ventricular contractions (PVCs) less than 6/min are common and not considered a concerning finding. Three or more consecutive PVCs at a rate of 100 bpm or more that terminate spontaneously within 30 seconds are considered nonsustained ventricular tachycardia (Marine, Card Electrophysiol Clin 2016, 8: 525-43; Buxton et al., Circulation 2006, 114: 2534-70).

29:

Following an initial 12-lead electrocardiogram (ECG) demonstrating no abnormal findings in the setting of hyperkalemia or hypokalemia, continued monitoring may be accomplished by telemetry (Bridgeman et al., Nephrol Dial Transplant 2019, 34: iii45-iii50).

30:

Infection and malfunction of diabetes technology (e.g., insulin pump), amongst others, are common precipitating causes of diabetic ketoacidosis (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; French et al., Bmj 2019, 365: l1114).

31:

National standards recommend diabetes self-management education and support (DSMES) for patients and/or caregivers to enhance knowledge, decision making, and mastery of skills for diabetes management to prevent acute complications, improve treatment outcomes, and reduce risk of long-term complications. Hospitalization represents a critical time to evaluate, assess, and provide DSMES that includes (American Diabetes Association, Diabetes Care 2024, 47: S1-S314):

- Identifying a health care provider to provide diabetes care post discharge
- Establishing home blood glucose goals, self-monitoring of blood glucose, and understanding when to call the provider
- Recognition, treatment, and prevention of hyperglycemia and hypoglycemia
- Administration of blood glucose-lowering medications, including insulin, if relevant
- Information on healthy dietary choices
- Proper use and disposal of insulin needles and syringes
- Sick-day management (e.g., when sick with a cold, flu, or infection)

32:

Critical events such as a new diagnosis, unmet treatment goals, complications, and transitions in life require evaluation of diabetes self-management educational needs to encourage acquiring the skills that may assist in executing and adhering to the treatment regimen and medical nutrition therapy (MNT) plan. An individualized MNT plan created through a joint effort of shared decision making with the patient and the healthcare team that includes a registered dietitian nutritionist is essential. The goals of MNT are to attain individualized glycemic targets, delay or prevent complications, reach and maintain body weight goals, incorporate personal and cultural preferences, address barriers to behavioral changes and access to healthy foods, and provide the tools and resources to develop and maintain healthy dietary patterns. Individualized macronutrient recommendations consider the patient's eating patterns, preferences, and metabolic goals, as evidence shows there are no general recommendations for macronutrients for all people with diabetes. Carbohydrate monitoring strategies are essential to achieve glycemic control and include, but are not limited to, carbohydrate counting, an exchange system, or experience-based estimation (American Diabetes Association, Diabetes Care 2024, 47: S1-S314; American Diabetes

Association, Diabetes Care 2022, 45:S1-S262).

33:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

34:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

35:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2 . The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

36:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

37:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin

North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

38:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

39:

Reference: (Jordan and Caesar, BMJ Qual Improv Rep 2015, 4:)

40:

The "R on T phenomenon" is the superimposition of an ectopic beat on the T wave of a preceding beat. Early observations suggest that R on T is likely to initiate sustained ventricular tachyarrhythmias.

41:

When T and U waves fuse, they form a TU fusion wave on an electrocardiogram. Severe hypokalemia may be associated with a large U wave created by T and U wave fusion (Zieg et al., Acta Paediatr 2016, 105: 762-72; Levis, Perm J 2012, 16: 53).

42:

The U wave occasionally follows the T wave on an electrocardiogram. A prominent upright U wave is associated with hypokalemia. When the U wave exceeds the T wave amplitude, the serum potassium level is less than 3 mEq/L. Severe hypokalemia may be associated with a giant U wave created by T and U wave fusion (Sandau et al., Circulation 2017, 136: e273-e344; Zieg et al., Acta Paediatr 2016, 105: 762-72).

43:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

44:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

45:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

46:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

47:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

48:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

49:

Cardiovascular disease (CVD) is the major cause of mortality and morbidity in patients with diabetes mellitus (DM). A blood pressure level less than 130/80 mmHg is recommended in individuals with diabetes to reduce CVD mortality. Patients with DM should be educated about lifestyle modifications that effectively reduce hypertension and dyslipidemia. These modifications may include (American Diabetes Association, Diabetes Care 2024, 47: S1-S314):

- Weight loss, if overweight
- Dietary Approaches to Stop Hypertension (DASH), including reducing sodium and increasing potassium intake
- Reduction of saturated fat, trans fat, and cholesterol consumption
- Increase of n-3 fatty acids, viscous fiber, and plant stanols/sterols consumption
- Moderation of alcohol consumption
- Increase in physical activity

In addition to lifestyle therapy, statin therapy is recommended for those with established CVD disease and risk factors. Moderate-intensity statin therapy is recommended for individuals with diabetes aged 40-75 years without CVD. High-intensity statin therapy should be initiated in people with diabetes aged 40-75 with one or more CVD risk factors, to achieve an LDL cholesterol less than 70 mg/dL and reduce LDL cholesterol by 50%(0.50) from baseline. Treatment recommendations for individuals with diabetes and established CVD include high-intensity statin therapy to lower LDL cholesterol by 50%(0.50) from baseline and an LDL cholesterol goal of less than 55mg/dL. Individuals with diabetes age 75 and older already on statin therapy can continue statin treatment. It may also be reasonable to initiate moderate-intensity statin therapy in adults with diabetes aged 75 and older after discussing the potential benefits and risks (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

50:

Large prospective randomized studies have shown that intensive diabetes management to achieve near-normal glucose levels delays the onset of nephropathy and diabetic retinopathy. Improved glycemic control has also been shown to modestly slow the progression of neuropathy. Reducing hypertension is also an effective way of preventing the progression or development of nephropathy and retinopathy (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

51:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

52:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

53:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

54:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

55:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

56:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

57:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit

- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

58:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

59:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

60:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

61:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

62:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

63:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

64:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

65:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

66:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

67:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

68:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784).

Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.